

Dynamic Receptive Fields in Mouse Auditory Cortex

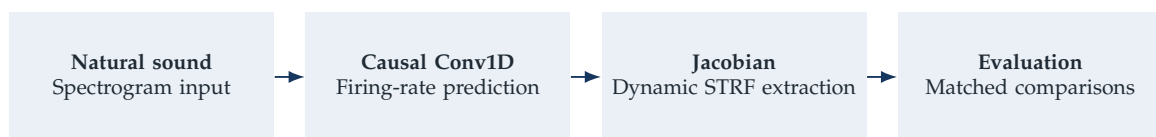
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Public technical summary. This project tested whether a deep neural model could predict auditory-cortical activity from natural ultrasonic vocalisations and expose response structure that static receptive-field methods miss.

Research question

Can a stimulus-dependent receptive field capture nonlinear neural-response behaviour that is not represented by static methods?

Analysis workflow



Method

- Trained a five-layer causal Conv1D model in PyTorch on sound spectrograms from 46 mouse auditory-cortex units.
- Used Jacobian-based automatic differentiation to derive dynamic spectrotemporal receptive fields from the trained network.
- Benchmarked against spike-triggered average, spike-triggered covariance, and maximum-noise-entropy models.

Evaluation

- Measured held-out prediction quality with Pearson correlation and noise ceilings.
- Used shuffle tests and matched comparison protocols to assess whether performance differences were robust.
- Examined temporal hold: persistence of a receptive-field pattern through time.

Evidence at a glance

PREDICTIVE PERFORMANCE

$$r = 0.339$$

Median held-out Pearson correlation for the dynamic STRF model.

TEMPORAL HOLD

43.5%

of analysed neurons showed a receptive-field pattern persisting over time.

Interpretation

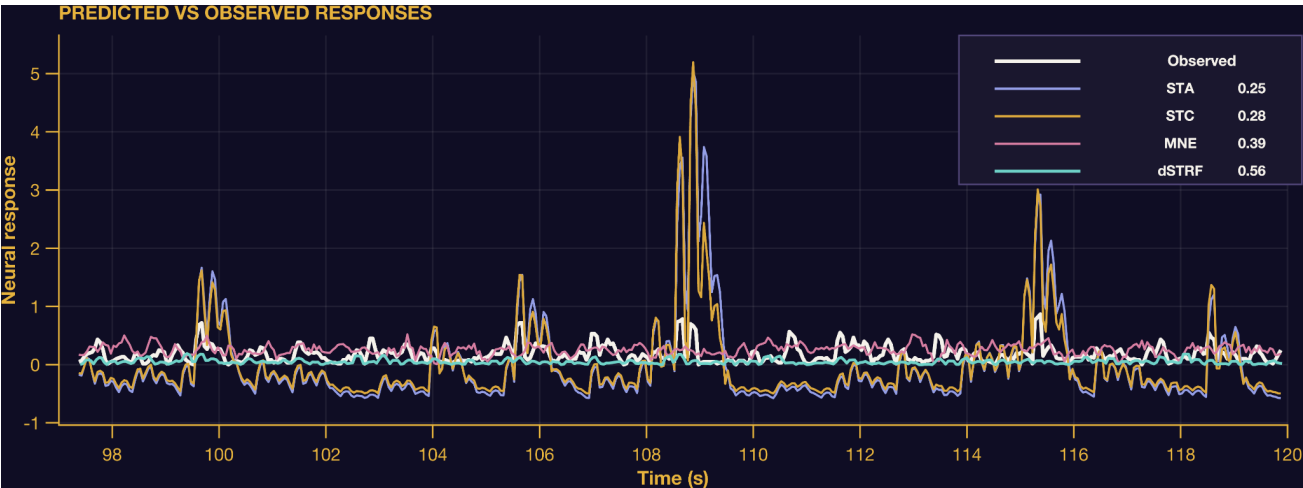
Under matched settings, the dynamic model outperformed static alternatives. The analysis also showed that temporal resolution and evaluation choices can materially alter conclusions. The resulting workflow supports context-dependent receptive-field analysis for naturalistic neural stimulation.

Selected Results

Dynamic receptive-field modelling in mouse auditory cortex

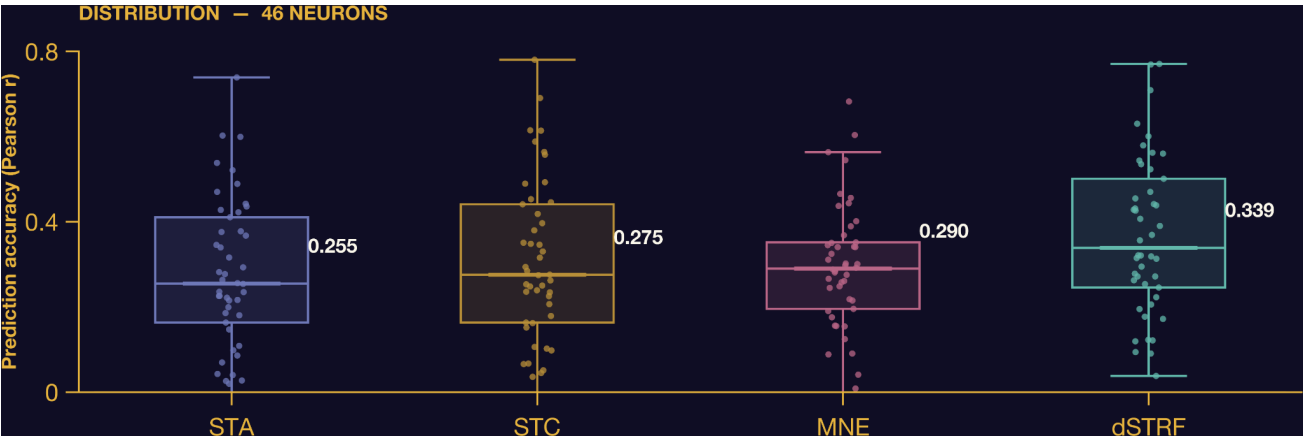
Interpretation. The dynamic STRF achieved the strongest median held-out prediction accuracy across the 46-unit population, while the trace below shows how the models differ on a representative response segment.

Predicted versus observed responses



A representative segment comparing observed neural activity with predictions from STA, STC, MNE, and dSTRF.

Prediction accuracy across 46 neurons



Boxplots show the distribution of Pearson correlations. The dSTRF model achieved the highest median prediction accuracy.